pensive. But here's the secret sauce: Offering fantastic after-sales service. It guarantees they will keep coming back for more of your awesome stuff and related goodies.

Even though it might not come inside the product package, if you want your product to stand out from the crowd, you have to make sure top-notch after-sales service is part of the whole deal.

Packaging consideration of charge controller

Let us explore how we choose the perfect packaging for our charge controller, but let us keep it simple! First, we need to consider the controller's size: It measures 270mm × 170mm × 80mm and weighs about 750 grams. There is nothing fragile inside, but we want to protect it from bending when handled.

Our customers are environmentally conscious, so we are committed to being eco-friendly. That is why we won't use any plastic in our packaging. Instead, we will use strong corrugated cardboard boxes. To prevent bending, we will use double-walled boxes and insert crushable cardboard pieces to keep the controller safely away from the box walls.

We also know that our product won't fly off the shelves right away. So, we need to think about shipping efficiency. The smallest container available is a 305cm (10-foot) container (ISO:668 D1) with dimensions of 2,802mm × 2,197mm × 2,330mm.

With the padding and reinforcements, the retail box will be 292mm × 192mm × 96mm. We will pack these smaller boxes into larger shipping cartons and then into the shipping container, limiting our stacking options.

There are six ways we can arrange the small boxes in the big container, and we have calculated how many boxes fit in each configuration (see Table 2). We will choose the configuration that allows us to pack the most efficiently and then design the shipping carton accordingly. Keep in mind that the final packed figure might be slightly different.

Our retail packaging will feature high-quality product images, specs, and even connection diagrams and installation instructions on the back. The charge controller will be set up with its HTML-based interface. We have made it even more convenient by including user registration, user manuals, and service manuals right on the web page.

For added assurance, all this information is available on our company's website. That way, customers can access it if anything goes wrong with the charge controller's embedded web server. Online manuals are a smart choice. They are eco-friendly, hard to lose, and easy to search. In addition, they can include audio-visual presentations when needed.

Final review

We have now reached the end of our series about systematic product development. Our journey began by exploring what makes a product truly lovable. From there, we delved into the QFD process, which helps gather customer and stakeholder expectations and turns them into specifications, design ideas, manufacturing plans, and quality assurance strategies.

We also took a quick look at concurrent engineering, a method that brings together team members from different areas to improve our designs efficiently. In one of the articles, we discussed cost optimisation for boosting profits, and in another, we focused on design risk analysis.

Next, we explored ways to make designs more manufacturing-friendly and less complex. Our seventh article delved into user interface and product styling, emphasising intuitive and user-friendly products through axiomatic design principles. We wrapped up our journey by discussing product strategy, including designing for future expansion and building a product family.

With this final article on product packaging, we hope our series has provided a glimpse into the various aspects of systematic product development. Hope these articles will offer guidance to aspiring entrepreneurs looking to create innovative products and contribute to the Make in India initiative. EFY